Stochastic optimization
Stochastic gradient descent

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Motivation

The recent years have seen a huge success of machine learning, and a renewed interest in the stochastic gradient algorithm, and the development of numerous variants.

The stochastic gradient algorithm is a special case of the stochastic approximation method, which was first introduced in 1951, and can be seen as an alternative to the sample-path approach.

The first part of these slides is based on S. Bhatnagar, H.L. Prasad, L.A. Prashanth, “Stochastic Recursive Algorithms for Optimization Simultaneous Perturbation Methods”, Springer-Verlag, 2013.

Robbins–Monro algorithm

Introduced in 1951, initially as a root-finding problem.

It extends easily to unconstrained optimization, the first-order necessary condition for

$$\min_{x \in \mathbb{R}^d} f(x) \qquad \text{being} \qquad \nabla_x f(x) = 0.$$

We therefore search for a zero of the gradient of $f(x)$.

We can also restrict the feasible domain to $\Theta \subseteq \mathbb{R}^d$:

$$\min_{x \in \Theta} f(x) = \mathbb{E}[Y(x, \xi)]$$

In this case, we have to assume that f reaches its minimum in the interior of Θ.

Problem in expectation

We consider here

$$f(x) = \mathbb{E}[Y(x, \xi)]$$

where the support of ξ is Ξ.

The problem to solve is

$$\nabla_x \mathbb{E}[Y(x, \xi)] = 0$$

We assume here that we can exchange the expectation and differentiation operators, i.e.

$$\nabla_x \mathbb{E}[Y(x, \xi)] = \mathbb{E}[\nabla_x Y(x, \xi)]$$

Stochastic approximation

Also known as the stochastic gradient descent (SGD) method.

Choose a starting point x_1.
$k \leftarrow 1$
while Stopping criteria not satisfied **do**
  Draw ξ_k from $\boldsymbol{\xi}$.
  Select a step length α_k.
  Compute

$$x_{k+1} = x_k - \alpha_k \nabla_x Y(x_k, \xi_k).$$

  $k \leftarrow k + 1$
end while

α_k is also called the learning rate.

Assumptions

Assume a unique minimizer x^*, and

A.1 $f(x)$ is continuously differentiable and its gradient is Lipschitz continuous with Lipschitz constant $L > 0$, i.e. $\forall\, x, y \in \mathbb{R}^d$,

$$\|\nabla_x f(x) - \nabla_x f(y)\|_2 \leq L\|x - y\|_2 \tag{A.1}$$

A.2 The iterates remain a.s. bounded, i.e.

$$\sup_k \|x_k\| < \infty \text{ almost surely.} \tag{A.2}$$

A.3 There exist scalars $M \geq 0$ and $M_V \geq 0$ s.t. $\forall\, k$,

$$\mathrm{Var}[\nabla_x Y(x, \xi_k)] \leq M + M_V \|\nabla_x f(x_k)\|_2^2 \tag{A.3}$$

A.4 The sequence α_k, $k = 1, 2, \ldots$, satisfies

$$\sum_{k=0}^{\infty} \alpha_k = \infty \quad \text{and} \quad \sum_{k=0}^{\infty} \alpha_k^2 < \infty. \tag{A.4}$$

Assumptions: notes

- In A.3, $\mathrm{Var}_{\xi_k}[\nabla_x Y(x, \xi_k)]$ does not refer to the covariance matrix of $\nabla_x Y(x, \xi_k)$.
- Variance of a random vector $g(\xi_k)$:

$$\begin{aligned}\mathrm{Var}_{\xi_k}[g] &= \mathbb{E}_{\xi_k}\left[\left\|g - \mathbb{E}_{\xi_k}[g]\right\|^2\right] \\ &= \mathbb{E}_{\xi_k}\left[\|g\|^2\right] - \left\|\mathbb{E}_{\xi_k}[g]\right\|^2.\end{aligned}$$

- A well-known consequence of the Lipschitz continuity assumption A.1 is

$$f(x) \leq f(y) + \nabla f(y)^T (x - y) + \frac{L}{2}\|x - y\|_2^2,$$

$\forall\, x, y \in \mathbb{R}^d$.

Step lengths

Consider the sequence of step lengths, also called positive gains sequence $\{\alpha_k \,|\, k \geq 1\}$.

This sequence satisfies the previous assumption in particular with

- $\alpha_k = \alpha/k$, given $\alpha > 0$.
- $\alpha_k = \alpha/k^\beta$, $\forall\, k \geq 1$, given $\alpha > 0$ and $\beta \in (0.5, 1)$.
- $\alpha_k = \alpha(\ln k)/k$, $\forall\, k \geq 2$, given $\alpha_1 = \alpha > 0$.
- $\alpha_k = \alpha/(k \ln k)$, $\forall\, k \geq 2$, given $\alpha_1 = \alpha > 0$.

Properties

- Very cheap iteration: gradient w.r.t. just one observation. No function evaluation.
- Reminder: d is a descent direction for f at x if

$$d^T \nabla_x f(x) < 0$$

- SGD is not a descent method as we can have $-\nabla_x Y(x, \xi_i)^T \mathbb{E}[\nabla_x Y(x, \xi)] \geq 0$ with $\nabla_x f(x) \neq 0$.
- Descent in expectation: if $\nabla_x f(x) \neq 0$,

$$\begin{aligned} &\mathbb{E}[-\nabla_x Y(x, \xi_i)^T \mathbb{E}[\nabla_x Y(x, \xi)]] \\ &= -\nabla_x \mathbb{E}[Y(x, \xi_i)]^T \nabla_x \mathbb{E}[Y(x, \xi)] \\ &= -\nabla_x f(x)^T \nabla_x f(x) < 0 \end{aligned}$$

Mini-batch method

Replace $\nabla_x Y(x, \xi_i)$ by

$$\frac{1}{n_k} \sum_{i=1}^{n_k} \nabla_x Y(x, \xi_i).$$

At each iteration, we take n_k new draws.

The cost per iteration is n_k times bigger, but

- it is a better estimate of the gradient
- the computation of the mini-batch can exploit parallelism

Batch method

Assume for now that the support Ξ is finite, of cardinality n. Then

$$\mathbb{E}[Y(x,\xi)] = \frac{1}{n}\sum_{i=1}^{n} Y(x,\xi_i), \quad \mathbb{E}[\nabla_x Y(x,\xi)] = \frac{1}{n}\sum_{i=1}^{n} \nabla_x Y(x,\xi_i)$$

Batch method:

$$x_{k+1} = x_k - \alpha_k \frac{1}{n}\sum_{i=1}^{n} \nabla_x Y(x,\xi_i).$$

In other words, we use all the observations to compute the true gradient.

Often, n is very large, and we prefer to work with $n_k << n$.

Stochastic approximation

We can generalize the expression of the stochastic approximation iteration using an estimator of the gradient of f at x_k:

$$x_{k+1} = x_k - \alpha_k \nabla \hat{f}(x_k).$$

As before, the gradient estimator can usually be taken as $\nabla Y(x_k, \xi_k)$, where $(\xi_k,\ k \geq 1)$ are i.i.d.

In that case, if $f(\cdot)$ is smooth, has a unique global minimizer x^*, and $\alpha_k = \alpha/k$ with $\alpha > 0$ sufficiently large, then under additional nonrestrictive conditions,

$$\sqrt{n}\,(x_n - x^*) \Rightarrow N(0, \Lambda),$$

as $n \to \infty$, for a certain $d \times d$ matrix Λ.

Convergence speed: stochastic boundedness

Source:
`https://en.wikipedia.org/wiki/Big_O_in_probability_notation`

We would like to measure how fast we converge to the solution, knowing that we generate a sequence of realizations of random variables.

Stochastic boundedness
The notation

$$X_n = O_p(a_n),$$

means that the set of values X_n/a_n is stochastically bounded:

$$\forall\, \epsilon > 0,\ \exists\, M > 0, N > 0 \text{ such that } P[|X_n/a_n| > M] < \epsilon\ \forall\, n > N.$$

$O_p(\cdot)$ vs $o_p(\cdot)$

Thus, $X_n = O_p(1)$ iff

$$\forall\, \epsilon > 0,\ \exists N_\epsilon, \delta_\epsilon \text{ such that } P[|X_n| \geq \delta_\epsilon] \leq \epsilon\ \forall\, n > N_\epsilon.$$

Convergence in probability

$X_n = o_p(1)$ iff

$$\forall\, \epsilon > 0, \delta > 0\ \exists N_{\epsilon,\delta} \text{ such that } P[|X_n| \geq \delta] \leq \epsilon\ \forall n > N_{\epsilon,\delta}.$$

Therefore $X_n = o_p(1) \Rightarrow X_n = O_p(1)$; the reverse does not hold.
More generally, $X_n = o_p(a_n)$ iff $X_n/a_n = o_p(1)$, i.e.

$$\forall\, \epsilon > 0 \lim_{n\to\infty} P[|X_n/a_n| \geq \epsilon] = 0.$$

Complexity

Source: Kim, Pasupathy, and Henderson, “A Guide to Sample Average Approximation”, in “Handbook of Simulation Optimization”, edited by Michael C. Fu, Springer, 2015.

Suppose the number of iterations completed in c units of computer time, $n(c)$, grows roughly linearly in c — as it would if sample gradients were computed in constant time.
A time-changed version of the CLT establishes that the resulting SA estimator has an error

$$x_{n(c)} - x^* = O_p(c^{-1/2}).$$

Equivalently, the computational effort required to obtain an error of order ϵ with SA is $O_p(\epsilon^{-2})$.

The performance of the recursion is highly dependent on the gain sequence $\{\alpha_n\}$.

Polyak–Ruppert averaging

Within the context of the SA iterative scheme, the fastest achievable convergence rate is $O_p(c^{-1/2})$.

This rate can be achieved under the "Polyak–Ruppert averaging".

- step-size sequence: $a_n = a/n^\gamma$ for some $\gamma \in (0, 1)$
- estimator of x^*:

$$\overline{x}_n = \frac{1}{n} \sum_{i=1}^{n} x_i.$$

Under mild conditions, the Polyak–Ruppert averaging scheme enjoys a CLT, although with a different covariance matrix Λ.

This happens irrespective of the value of the constant $a > 0$ (but the choice of a affects the small-sample performance). The Polyak–Ruppert averaging scheme also has other optimality properties related to the matrix Λ.

Order of Convergence

Denote the numerical procedure acting on the sample function $f_n(x)$ by the mapping $A(x) : \Theta \to \Theta$.

Let $A_k(x)$ represent the iterate obtained after k successive applications of $A(\cdot)$ on the initial iterate x.

Assume that the function $f_n(x)$ attains its infimum $v_n^* := \inf\{f_n(x) : x \in \Theta\}$ and that $f_n(A_k(x)) \to v_n^*$ as $k \to \infty$ for all $x \in \Theta$. Also, to avoid trivialities, assume that $f_n(A_{k+1}(x))$ is different from v_n^* for all k.

Sublinear convergence

Denote

$$Q_t = \lim \sup_{k \to \infty} \frac{|f_n(A_{k+1}(x)) - v_n^*|}{|f_n(A_k(x)) - v_n^*|^t}.$$

Definition

$A(x) : \Theta \to \Theta$ is said to exhibit p^{th}-order sublinear convergence if $Q_1 \geq 1$, and

$$\exists\, p, s > 0 \text{ such that } p = sup\{r : f_n(A_k(x)) - v_n^* \leq s/k^r,\ \forall\, x \in \Theta\}.$$

Linear convergence

Definition

The numerical procedure $A(x) : \Theta \to \Theta$ is said to exhibit linear convergence if $Q_1 \in (0, 1)$ for all $x \in \Theta$.

The definition of linear convergence implies that there exists a constant β satisfying $f_n(A(x)) - v_n^* \leq \beta(f_n(x) - v_n^*)$ for all $x \in \Theta$. The projected gradient method with Armijo steps when executed on certain smooth problems exhibits a linear convergence rate.

Superlinear convergence

Definition

The numerical procedure $A(x) : \Theta \to \Theta$ is said to exhibit superlinear convergence if $Q_1 = 0$ for all $x \in \Theta$. The convergence is said to be p^{th}-order superlinear if $Q_1 = 0$ and $\sup\{t : Q_t = 0\} = p < \infty$ for all $x \in \Theta$.

When $f_n(x)$ is strongly convex and twice Lipschitz continuously differentiable with observable derivatives, Newton method is second-order superlinear. For settings where the derivative is unobservable, there is a slight degradation in the convergence rate, but Newton method remains superlinear.

Convergence rate for the SAA method

Theorem

Assumptions:

1. $\mathbb{E}[Y^2(x,\xi)] < \infty$ *for all* $x \in \Theta$.
2. *The function* $Y(x,\xi)$ *is Lipschitz w.p.1, with Lipschitz constant* $K(\xi)$*, and* $\mathbb{E}[K(\xi)] < \infty$.
3. *The function* $f_n(x)$ *attains its infimum on* Θ *for each* n *w.p.1.*

Let $c = n \times k$ *and* $n/c^{1/(2p+1)} \to a$ *as* $c \to \infty$*, with* $a \in (0,\infty)$*. Then, if the numerical procedure exhibits* p^{th}*-order sublinear convergence,*

$$c^{p/(2p+1)} \left(f_n(A^k(x)) - v^* \right) = O_p(1)$$

as $c \to \infty$.

Convergence rate for the SAA method

Main insight: the maximum achievable convergence rate with the SAA method, is $O_p(c^{-p/(2p+1)})$ when the numerical procedure in use exhibits p^{th}-order sublinear convergence.

It is also possible to show that the corresponding rates when using linearly convergent and p^{th}-order superlinearly convergent procedures are $O_p((c/\log c)^{-1/2})$ and $O_p((c/\log\log c)^{-1/2})$, respectively.

None of the families of numerical procedures considered are capable of attaining the canonical convergence rate $O_p(c^{-1/2})$.

The generic SG method

Source:

- Léon Bottou, Frank E. Curtis, and Jorge Nocedal, Optimization Methods for Large-Scale Machine Learning, SIAM Review 60(2), 2018, pp. 223–311, https://doi.org/10.1137/16M1080173
- Léon Bottou, Frank E. Curtis, and Jorge Nocedal, Optimization Methods for Machine Learning Part II – The theory of SG, https://icml.cc/Conferences/2016/tutorials/part-2.pdf

We generalize the stochastic gradient method with the update

$$x_{k+1} = x_k - \alpha_k g(x_k, \xi_k).$$

instead of

$$x_{k+1} = x_k - \alpha_k \nabla_x Y(x_k, \xi_k).$$

The generic SG method

The function $f : \mathbb{R}^d \to \mathbb{R}$ could be

$$f(x) = \begin{cases} R(x) = \mathbb{E}[Y(x;\boldsymbol{\xi})] & \text{the expected risk,} \\ R_n(x) = \frac{1}{n}\sum_{\xi=1}^{n} Y(x;\xi) & \text{the empirical risk.} \end{cases}$$

The stochastic vector could be

$$g(x;\xi_k) = \begin{cases} \nabla_x Y(x_k,\xi_k) & \text{(one realization)} \\ \frac{1}{n_k}\sum_{i=1}^{n_k} \nabla_x Y(x_k,\xi_k) & \text{(minibatch)} \\ B_k \frac{1}{n_k}\sum_{i=1}^{n_k} \nabla_x Y(x_k,\xi_k) & \text{(rescaled minibatch)} \end{cases}$$

Stochastic processes

While we assume the draws ξ_i, $i = 1, 2, \ldots$ are i.i.d., it is possible to extend the results to the situation where $\{\xi_i, i = 1, 2, \ldots\}$ form an adapted stochastic process, where each ξ_i can depend on the previous ones.

Active learning

- In active learning, $g(x_k; \xi_k)$ produces a multinomial distribution on the training examples in a manner that depends on the current solution x_k.
- ξ_k is then transformed to draw from this distribution.

Active learning is not covered here, but again, the results can be extended to this situation.

Smoothness

Theorem

Under Assumption A.1 (Lipschitz continuity), $\forall k \in \mathbb{N}$, the iterates of the SG method satisfy

$$\mathbb{E}_{\xi_k}[f(x_{k+1})] - f(x_k) \leq -\alpha_k \nabla f(x_k)^T \mathbb{E}_{\xi_k}[g(x_k, \xi_k)] + \frac{1}{2}\alpha_k^2 L \mathbb{E}_{\xi_k}\left[\|g(x_k, \xi_k)\|_2^2\right].$$

- $\alpha_k \nabla f(x_k)^T \mathbb{E}_{\xi_k}[g(x_k, \xi_k)]$: expected decrease;
- $\frac{1}{2}\alpha_k^2 L \mathbb{E}_{\xi_k}\left[\|g(x_k, \xi_k)\|_2^2\right]$: noise.

Smoothness: proof

From A.1, we have

$$f(x_{k+1}) - f(x_k) \leq \nabla f(x_k)^T (x_{k+1} - x_k) + \frac{L}{2} \|x_{k+1} - x_k\|_2^2.$$

Since $x_{k+1} = x_k - \alpha_k g(x_k, \xi_k)$, this leads to

$$f(x_{k+1}) - f(x_k) \leq -\alpha_k \nabla f(x_k)^T g(x_k, \xi_k) + \alpha_k^2 \frac{L}{2} \|g(x_k, \xi_k)\|_2^2.$$

This implies

$$\mathbb{E}_{\xi_k}[f(x_{k+1}) - f(x_k)] \leq \mathbb{E}_{\xi_k}\left[-\alpha_k \nabla f(x_k)^T g(x_k, \xi_k) + \alpha_k^2 \frac{L}{2} \|g(x_k, \xi_k)\|_2^2\right]$$

or

$$\mathbb{E}_{\xi_k}[f(x_{k+1})] - f(x_k) \leq -\alpha_k \nabla f(x_k)^T \mathbb{E}_{\xi_k}[g(x_k, \xi_k)] + \alpha_k^2 \frac{L}{2} \mathbb{E}_{\xi_k}\left[\|g(x_k, \xi_k)\|_2^2\right].$$

Assumptions: first and second moment limits

The SG method applied to $f(\cdot)$ satisfies

a) There exists an open set $\mathcal{A}$ such that

$$f(x) \geq f_{lb}(x), \ \forall\, x \in \mathcal{A},$$

and

$$x_k \in \mathcal{A}, \ \forall k \in \mathbb{N} \tag{A.5}$$

b) $\exists \mu, \mu_G$ such that $0 < \mu < \mu_G$ and $\forall k \in \mathbb{N}$,

$$\begin{aligned} \nabla f(x_k)^T \mathbb{E}_{\xi_k}[g(x_k, \xi_k)] &\geq \mu \|\nabla f(x_k)\|_2^2, \\ \|\mathbb{E}_{\xi_k}[g(x_k, \xi_k)]\|_2 &\leq \mu_G \|\nabla f(x_k)\|_2. \end{aligned} \tag{A.6}$$

c) $\exists M \geq 0, M_V \geq 0$ such that $\forall k \in \mathbb{N}$,

$$\begin{aligned} \mathrm{Var}_{\xi_k}[g(x_k, \xi_k)] &= \mathbb{E}_{\xi_k}\left[\|g(x_k, \xi_k)\|_2^2\right] - \left\|\mathbb{E}_{\xi_k}[g(x_k, \xi_k)]\right\|_2^2 \\ &\leq M + M_V \|\nabla f(x_k)\|_2^2. \end{aligned} \tag{A.7}$$

Assumptions: notes

- A.6 expresses that in expectation, $g(x_k, \xi_k)$ is a sufficient descent direction.
  - True if $\mathbb{E}_{\xi_k}[g(x_k, \xi_k)] = H_k \nabla f(x_k)$ with H_k positive definite and bounded spectrum.
  - Particular case: $H_k = I$. Then A.6 holds with $\mu = \mu_G = 1$.
- A.7 is a direct generalization of A.3.
- From A.6 and A.7,

$$\begin{aligned}\mathbb{E}_{\xi_k}\left[\|g(x_k, \xi_k)\|_2^2\right] &\leq \left\|\mathbb{E}_{\xi_k}[g(x_k, \xi_k)]\right\|_2^2 + M + M_V \|\nabla f(x_k)\|_2^2 \\ &\leq \mu_G^2 \|\nabla f(x_k)\|_2^2 + M + M_V \|\nabla f(x_k)\|_2^2 \\ &= M + M_G \|\nabla f(x_k)\|_2^2,\end{aligned}$$

with $M_G = M_V + \mu_G^2 \geq \mu^2 > 0$.

Moments

Theorem (Moments)

Under Assumptions A.1 and A.5–A.7, $\forall k \in \mathbb{N}$,

$$\begin{aligned}\mathbb{E}_{\xi_k}[f(x_{k+1})] - f(x_k) &\leq -\mu\alpha_k \|\nabla f(x_k)\|^2 + \frac{1}{2}\alpha_k^2 L \mathbb{E}_{\xi_k}\left[\|g(x_k, \xi_k)\|_2^2\right] \\ &\leq -\alpha_k \left(\mu - \frac{1}{2}\alpha_k L M_G\right) \|\nabla f(x_k)\|_2^2 + \frac{1}{2}\alpha_k^2 L M.\end{aligned}$$

- $\left(\mu - \frac{1}{2}\alpha_k L M_G\right) \|\nabla f(x_k)\|_2^2$: expected decrease;
- $\frac{1}{2}\alpha_k^2 L M$: noise.

Proof

We already proved

$$\mathbb{E}_{\xi_k}[f(x_{k+1})] - f(x_k) \leq -\alpha_k \nabla f(x_k)^T \mathbb{E}_{\xi_k}[g(x_k, \xi_k)] + \alpha_k^2 \frac{L}{2} \mathbb{E}_{\xi_k}\left[\|g(x_k, \xi_k)\|_2^2\right].$$

From A.6, this leads to

$$\mathbb{E}_{\xi_k}[f(x_{k+1})] - f(x_k) \leq -\alpha_k \mu \|\nabla f(x_k)\|_2^2 + \alpha_k^2 \frac{L}{2} \mathbb{E}_{\xi_k}\left[\|g(x_k, \xi_k)\|_2^2\right],$$

giving the first inequality. Since

$$\mathbb{E}_{\xi_k}\left[\|g(x_k, \xi_k)\|_2^2\right] \leq M + M_G \|\nabla f(x_k)\|_2^2,$$

we have

$$\begin{aligned}\mathbb{E}_{\xi_k}[f(x_{k+1})] - f(x_k) &\leq -\alpha_k \mu \|\nabla f(x_k)\|_2^2 + \alpha_k^2 \frac{L}{2}\left(M + M_G \|\nabla f(x_k)\|_2^2\right) \\ &= -\alpha_k \left(\mu - \frac{1}{2}\alpha_k L M_G\right) \|\nabla f(x_k)\|_2^2 + \frac{1}{2}\alpha_k^2 L M.\end{aligned}$$

Strong convexity

Consider $f(\cdot) \in C^2$ with a convex domain.

- Convexity:

$$\forall x, y \in dom(f),\ f(y) \geq f(x) + \nabla f(x)^T (y - x)$$

- Strong convexity

$$\exists c > 0, \forall x, y \in dom(f),\ f(y) \geq f(x) + \nabla f(x)^T (y - x) + \frac{1}{2} c \|y - x\|_2^2. \quad \text{(A.8)}$$

  Alternate definition if $f(\cdot) \in C^2$:

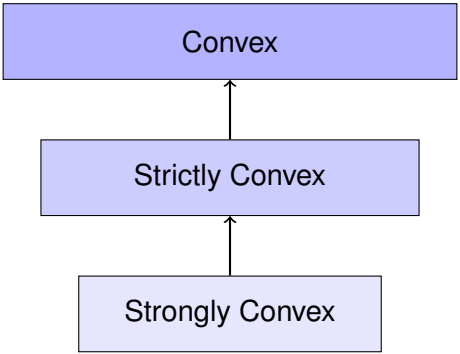
$$\nabla^2 f(x) \succeq cI,$$

  where I is the identity matrix.

Implications of strong convexity:

- $\exists \mu > 0, \forall x, y \in dom(f),\ \|\nabla_x f(x) - \nabla_x f(y)\|_2 \geq \mu \|x - y\|_2$
- $\forall x \in dom(f), 2c(f(x) - f^*) \leq \|\nabla f(x)\|_2^2$

Hierarchy of Convexity

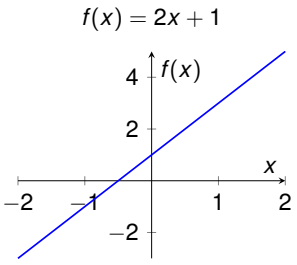


Implication chain:

$$\text{Strongly convex} \Rightarrow \text{Strictly convex} \Rightarrow \text{Convex}$$

The reverse implications do not hold!

Example 1: Linear Function

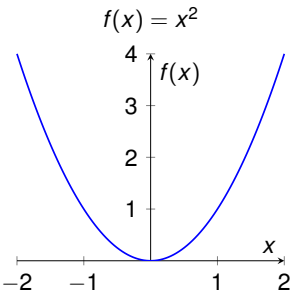


Classification:

- ✓ Convex
- × Strictly convex
- × Strongly convex

Linear functions satisfy convexity with equality, but not the strict inequalities.

Example 2: Quadratic Function

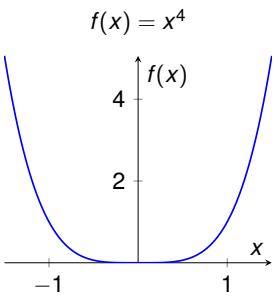


Classification:

- ✓ Convex
- ✓ Strictly convex
- ✓ Strongly convex ($m = 2$)

The second derivative $f''(x) = 2 > 0$ ensures strong convexity.

Example 3: Quartic Function

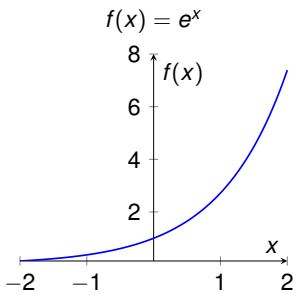


Classification:

- ✓ Convex
- ✓ Strictly convex
- $\times$ Strongly convex

$f''(0) = 0$ means no uniform lower bound on curvature. Strictly convex but not strongly convex.

Example 4: Exponential Function

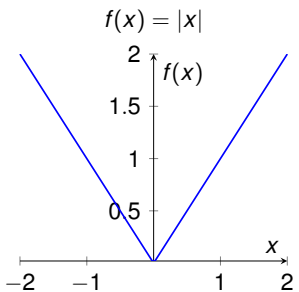


Classification:

- ✓ Convex
- ✓ Strictly convex
- × Strongly convex (globally)

Strictly convex since $f''(x) = e^x > 0$, but $f''(x) \to 0$ as $x \to -\infty$.

Example 5: Piecewise Linear Function



Classification:

- ✓ Convex
- × Strictly convex
- × Strongly convex

Convex but has linear segments where strict inequality fails.

Strong convexity

- Simplify the theoretical analysis and provide the strongest results.
- We can use regularization to locally convexify the function.
- Strong local minimizer: $\exists \epsilon > 0$ such that
  $f(x) > f(x^*) + \epsilon^* \|x - x^*\|_2, \ \forall \ x \in \mathcal{B}(x^*, \epsilon) \setminus \{x^*\}$.
  Therefore, a strong local minimizer is a strict local minimizer: $\exists \epsilon > 0$ such that
  $f(x) > f(x^*), \ \forall \ x \in \mathcal{B}(x^*, \epsilon) \setminus \{x^*\}$.
- If $f(x)$ is locally strongly convex in $\mathcal{B}(x^*)$, any local minimizer x^* is a strong local minimizer.
- But a strong local minimizer does not imply local strong convexity. Example: $f(x) = x^4$.

Total expectation

- $\mathbb{E}_k$: expectation w.r.t. ξ_k.
- $\mathbb{E}$: full expectation w.r.t. all $\xi_1, \ldots, \xi_k$:

$$\mathbb{E}[x_{k+1}] = \mathbb{E}_{\xi_1}[\mathbb{E}_{\xi_2}[\ldots \mathbb{E}_{\xi_k}[x_{k+1}]]]$$

SG with fixed stepsize

Theorem

Under Assumptions A.1, A.5–A.7, and A.8, if we apply the SGD with a fixed stepsize $\alpha_k = \overline{\alpha}$ satisfying

$$0 < \overline{\alpha} \leq \frac{\mu}{LM_G}, \tag{1}$$

then, for all $k \in \mathbb{N}$,

$$\mathbb{E}[f(x_k) - f^*] \leq \frac{\overline{\alpha}LM}{2c\mu} + (1 - \overline{\alpha}c\mu)^{k-1}\left(f(x_1) - f^* - \frac{\overline{\alpha}LM}{2c\mu}\right) \overset{k\to\infty}{\longrightarrow} \frac{\overline{\alpha}LM}{2c\mu}.$$

- Converges only to a neighborhood of optimum.

Proof

The moments theorem, with $\alpha_k = \overline{\alpha}$ and (1), gives

$$\begin{aligned}\mathbb{E}_{\xi_k}[f(x_{k+1})] - f(x_k) &\leq -\overline{\alpha}\left(\mu - \frac{1}{2}\overline{\alpha}LM_G\right)\|\nabla f(x_k)\|_2^2 + \frac{1}{2}\overline{\alpha}^2 LM \\ &\leq -\frac{1}{2}\overline{\alpha}\mu\|\nabla f(x_k)\|_2^2 + \frac{1}{2}\overline{\alpha}^2 LM.\end{aligned}$$

From strong convexity, this implies

$$\mathbb{E}_{\xi_k}[f(x_{k+1})] - f(x_k) \leq -\overline{\alpha}c\mu\,(f(x_k) - f^*) + \frac{1}{2}\overline{\alpha}^2 LM,$$

or equivalently

$$\mathbb{E}_{\xi_k}[f(x_{k+1})] - f^* \leq f(x_k) - f^* - \overline{\alpha}c\mu\,(f(x_k) - f^*) + \frac{1}{2}\overline{\alpha}^2 LM.$$

Proof

Thus,

$$\mathbb{E}\left[\mathbb{E}_{\xi_k}[f(x_{k+1})] - f^*\right] \leq (1 - \overline{\alpha} c\mu)\mathbb{E}\left[f(x_k) - f^*\right] + \frac{1}{2}\overline{\alpha}^2 LM,$$

or

$$\begin{aligned}
\mathbb{E}\left[f(x_{k+1}) - f^*\right] - \frac{\overline{\alpha} LM}{2c\mu} &\leq (1 - \overline{\alpha} c\mu)\left(\mathbb{E}\left[f(x_k) - f^*\right] - \frac{\overline{\alpha} LM}{2c\mu}\right) \\
&\leq (1 - \overline{\alpha} c\mu)^{k-1}\left(\mathbb{E}\left[f(x_1) - f^*\right] - \frac{\overline{\alpha} LM}{2c\mu}\right) \\
&\overset{k\to\infty}{\longrightarrow} \frac{\overline{\alpha} LM}{2c\mu},
\end{aligned}$$

as

$$0 < \overline{\alpha} c\mu \leq \frac{c\mu^2}{LM_G} \leq \frac{c\mu^2}{L\mu^2} \leq 1.$$

SG with fixed stepsize

- If noise $M = 0$, recovers linear convergence.
- If $M > 0$, noise limits accuracy.

Analysis

More details at:

- https://icml.cc/2016/tutorials/part-2.pdf
- https://icml.cc/2016/tutorials/part-3.pdf

We will refer to this material.